

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Class Test

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA15

---

### COURSE OUTCOME COVERED

**CO4:** *Analyze big data characteristics and challenges across domains including security and application aspects.*  
(BL4)

Dave's Taxonomy: Articulation (Level 4)  
Question Count: 10

**Total Marks: 30**

---

### Code of Conduct

I **SWETA R(192511408)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

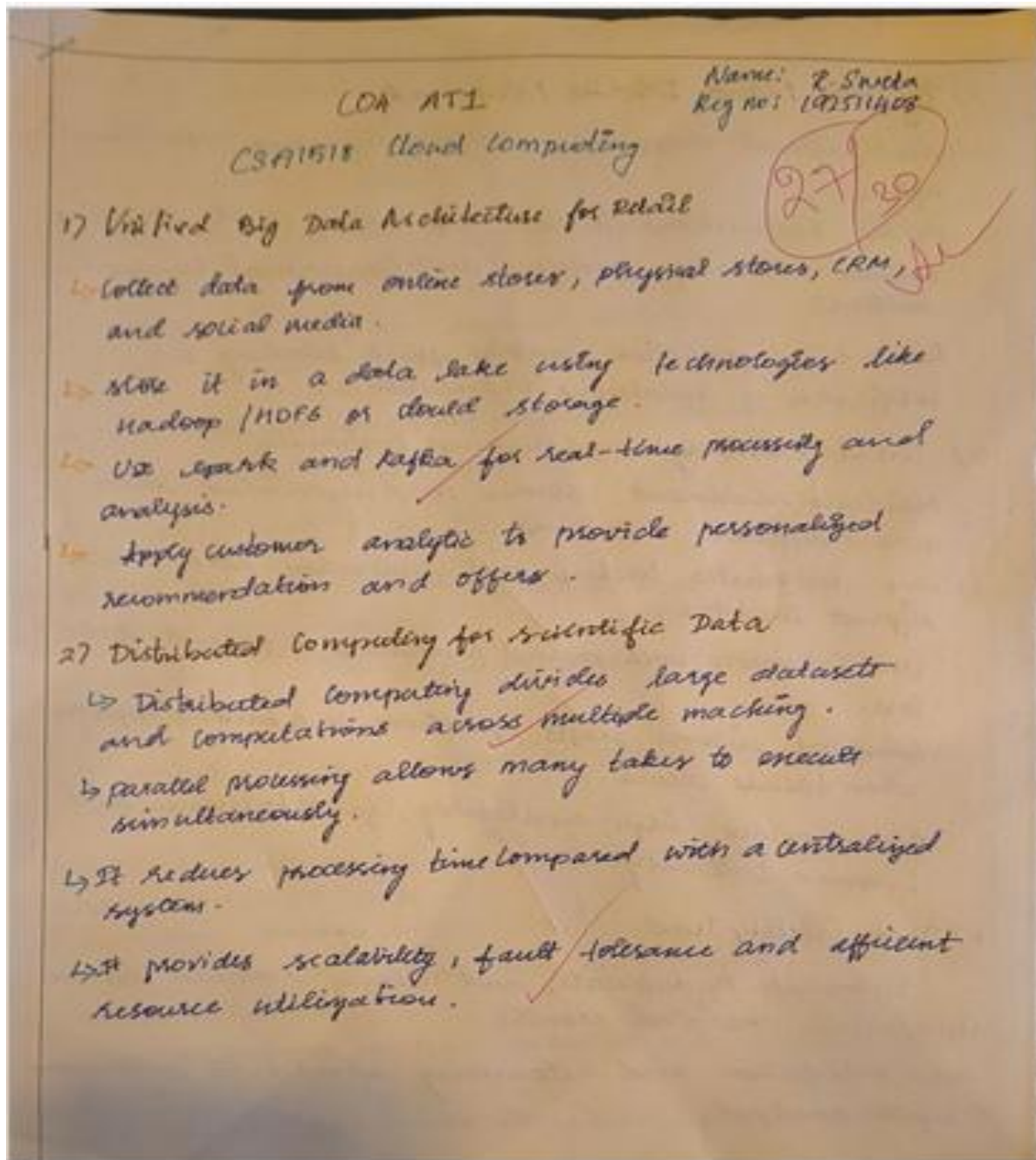
Signature of the student: SWETA R

### Assessment Tasks

#### Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.



### 3) Big Data Fraud Detection Framework

- Collect payment transactions continuously using Kafka/Streams.
- Process transactions in real time using Spark stream.
- Machine-learning models identify unusual transaction patterns.
- Real-time prediction enables quick blocking or verification of suspicious transactions.

### 5) Disaster Recovery and Business Continuity.

- Maintain redundant services and replicated data across different locations.
- Uses automatic backup and replication data across different locations.
- Use automatic backup and replication to prevent data loss.
- Automated failover shifts operations to a backup system when failure occurs.
- This provides high availability, fault tolerance and business continuity.

### 6) Data Quality Issue.

- Inconsistent, duplicate, and missing data can produce inaccurate analytical results.
- Data validation and cleansing should be performed before analysis.

## 7) Recommendation System.

- ↳ Collect user behavior such as views, searches, ratings, and watch history.
- ↳ Store and process large-scale data using cloud storage and Spark.
- ↳ Machine-learning models identify user preference and similar content.
- ↳ Personalized recommendations improve user engagement and satisfaction.

## 8) Cloud Computing in Big Data.

- ↳ Cloud provides scalable storage and computing resources on demand.
- ↳ It reduces initial infrastructure cost and provides flexibility.
- ↳ Resources can be increased or decreased according to workload.
- ↳ Challenges include security, privacy, vendor dependency, and unpredictable costs.
- ↳ **Multi Data management.**
  - Video, audio, and images are large, unstructured and storage intensive.
  - Use distributed storage such as HDFS or cloud computing objects.
  - Technologies like Spark can process multimedia data parallel.



#### 9) Realtime Transportation Analytics

- ↳ Collects GPS, traffic, fuel, and vehicle data using IoT sensors and streaming systems.
- ↳ Processes real-time data using Kafka and Spark streaming.
- ↳ Analytics can identify the fastest routes and avoid traffic congestion.
- ↳ This improves route optimization, fuel efficiency, and operational decisions.

#### 10) Ethical and Legal Challenges

- ↳ Big data can create issues involving privacy, unauthorized access, and misuse of personal data.
- ↳ Organizations must obtain appropriate consent and collect only necessary data.
- ↳ Use encryption, access control, anonymization, and regular audits.
- ↳ Following data protection law and transparent policies improve user trust and compliance.

Class Test Rubric

| Criteria             | Weightage | Level 4 - Exemplary                               | Level 3 - Proficient    | Level 2 - Developing | Level 1 - Beginning |
|----------------------|-----------|---------------------------------------------------|-------------------------|----------------------|---------------------|
| Concept Accuracy     | 30%       | Accurate and well-expressed technical answers     | Mostly accurate answers | Partial accuracy     | Inaccurate answers  |
| Coverage             | 20%       | Covers all required points                        | Covers most points      | Limited coverage     | Very poor coverage  |
| Use of Examples      | 15%       | Uses relevant and meaningful examples             | Some relevant examples  | Weak examples        | No useful examples  |
| Analytical Awareness | 20%       | Shows strong awareness of issues and implications | Reasonable awareness    | Basic awareness      | Minimal awareness   |

|         |     |                           |                 |                          |              |
|---------|-----|---------------------------|-----------------|--------------------------|--------------|
| Clarity | 15% | Clear and concise writing | Generally clear | Some unclear expressions | Poor clarity |
|---------|-----|---------------------------|-----------------|--------------------------|--------------|